G D HRUTHIK

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years of experience designing, deploying, and optimizing AI systems and ML pipelines in healthcare and enterprise domains. Expertise in Python, cloud computing, and MLOps, with a track record of building transparent, regulatory-compliant models that drive measurable impact. Recognized for enhancing AI adoption among scientists and transitioning models from research to production.

EDUCATION

New England College, Henniker, NH **Jun 2025**
Master of Science, Data Science and Analytics
- **GPA:** 3.74

PROFESSIONAL EXPERIENCE

Sanofi **Apr 2024 - Present**
Machine Learning Engineer *Boston, MA*
- Collaborated with clinical teams to design ML pipelines for genomic and clinical trial data, enhancing experiment reproducibility and ensuring transparency crucial for scientific accuracy.
- Containerized models using Docker and Kubernetes, enabling scalable deployment across oncology and immunology studies.
- Implemented model monitoring, versioning, and compliance checks aligned with FDA and GxP requirements, reinforcing regulatory adherence.
- Developed internal AI tools with Streamlit and FastAPI, increasing adoption of AI solutions among scientists by 40% through enhanced explainability and cross-functional collaboration.
- Integrated EHR and biomarker data into production pipelines to support translational medicine teams in advanced biomarker discovery.
- Engineered a computer vision workflow with CNNs and PyTorch for microscopy images, reducing protein misfolding detection time by 50% compared to manual review.

Cognizant Technology Services **Aug 2021 - May 2023**
Machine Learning Engineer *Coimbatore, TN, India*
- Conceived real-time anomaly detection models with Autoencoders and Isolation Forests, reducing industrial equipment downtime by 20% and ensuring data-driven decision-making.
- Constructed predictive maintenance pipelines using Random Forest, Kafka, and Azure Stream Analytics to forecast equipment failures with precision.
- Deployed a fraud detection solution in Azure ML that achieved inference speeds under 120ms and enhanced accuracy, demonstrating robust model optimization and reliability.
- Supported an e-commerce recommendation system in TensorFlow, contributing to a 15% increase in click-through rates by refining model performance.
- Coordinated on end-to-end ML operations pipelines, standardizing containerization and monitoring practices to streamline cross-functional AI development processes.

TECHNICAL SKILLS

- **Programming & Data Analysis**: Python, C++, R, SQL, Scala, Pandas, NumPy
- **Machine Learning**: Regression, Classification, Clustering, Dimensionality Reduction, Ensemble Methods (Bagging, Boosting, Stacking), Hyperparameter Tuning (Optuna, Grid Search)
- **Deep Learning & Computer Vision**: CNN, RNN, LSTM, BERT, CodonBert, ProGen2 Transformers, Attention Mechanisms, Reinforcement Learning, CLIP, Image Classification, Object Detection, Fine-tuning Models, Diffusion Models
- **Frameworks & Libraries**: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, DeepSpeed, Flask, Django
- **Natural Language Processing**: Text Preprocessing, Tokenization, Word Embeddings, Sentiment Analysis, LangChain, RAG, Hugging Face Transformers, OpenAl API
- **Generative AI**: GPT-based LLMs, Diffusion Models, Prompt Engineering
- **Vector Search & Retrieval**: FAISS, Pinecone, Chroma, Weaviate
- **MLOps & Model Deployment**: MLflow, DVC, Docker, Kubernetes, Azure Kubernetes Service (AKS), Streamlit, Ray, TorchServe, FastAPI, Triton Inference Server, TensorFlow Serving
- **CI/CD & Workflow Orchestration**: Jenkins, GitHub Actions, Apache Airflow, Azure DevOps, Luigi
- **Cloud Platforms**: AWS SageMaker, GCP Vertex Al, Azure ML Studio
- **Visualization & Reporting**: Tableau, Matplotlib, Seaborn
- **Data Engineering**: Hadoop, PySpark, Azure Databricks, Data Factory, Synapse Analytics, Snowflake
- **Project & Collaboration Tools**: Jupyter, Google Colab, VS Code, Agile, A/B Testing, Requirement Gathering